

Who Gets the Best Loan?

Visualizing Credit Risk Patterns in LendingClub Data

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Data Understanding

The dataset comes from LendingClub, a major US peer-to-peer lending platform. It contains 10,000 loan records spanning 2012 to 2018, each described by 91 attributes covering the borrower's financial profile, credit history, and loan repayment behavior.

The central variable is **grade**, an ordinal label from A (lowest risk) to G (highest risk) assigned by LendingClub to reflect how likely a borrower is to default. The grade distribution is skewed toward lower-risk loans: grades A and B make up nearly half the dataset, while F and G together account for less than 3%.

The core question driving this project is: what patterns in borrower behavior and financial health actually separate a grade-A loan from a grade-D one? And are those patterns consistent across states, years, and loan purposes? We hypothesize that higher-risk grades will be associated with higher debt-to-income ratios and lower FICO scores, and that certain loan purposes such as small business lending will skew toward lower grades regardless of income level.

Data Mining

All three files were provided directly as part of the assignment. The learning set (`loan-10k_lrn.csv`) contains both features and the target grade, and serves as the primary dataset for exploration and visualization. The test set (`loan-10k_test.csv`) holds the same features without labels, and the solution example (`loan-10k_sol_ex.csv`) defines the expected output format.

No external data retrieval is needed. The data is already clean at the file level and ready to load.

Data Cleaning

The dataset is relatively clean out of the box. There are no missing values in core financial columns. Some behavioral attributes such as `mths_since_recent_inq` or `mo_sin_old_il_acct` have sporadic gaps, which likely reflect genuinely absent credit history rather than data errors. These will be treated as a meaningful signal where relevant, and excluded from analyses where they would distort results. Numerical features with extreme outliers will be identified through distribution inspection and handled by capping or exclusion on a per-variable basis, and continuous features will be normalized where required by the modeling step.

Date fields have already been split into separate month and year columns. No merging of external datasets is planned.

Data Exploration

The exploration will focus on understanding what drives grade assignment across multiple dimensions.

For **financial risk factors**, scatterplots will show how interest rate, debt-to-income ratio, and revolving utilization relate to grade. These are quantitative attributes best explored with position-based encodings, since small differences in correlation matter here and area or angle would introduce perceptual error.

For **categorical breakdowns**, bar charts will compare grade distributions across loan purpose, employment length, and home ownership status. Bar charts are preferred over pie charts because length comparisons are more accurate than angle for this type of task.

For **geographic patterns**, a choropleth map over US states will show whether certain states cluster around higher or lower grades. This supports a spatial comparison task.

For **temporal trends**, a stacked bar chart over issue year will show how grade composition shifted from 2012 to 2018. We hypothesize that higher-risk grades (D through G) grew as a proportion of total loans after 2015, potentially reflecting platform-level expansion and looser underwriting criteria during a period of low default rates.

For **high-dimensional structure**, parallel coordinates will let us see how multiple financial attributes simultaneously differ across grade groups, even if clutter will need to be managed. Tableau's parallel coordinates custom visual will be used, with the global grade filter reducing visible lines to a manageable subset. The initial view will display only grade A and grade G as visual extremes, with intermediate grades added progressively.

Interactive features will be implemented as a linked Tableau dashboard. Each chart will be configured as a filter so that selecting a grade bar, a state on the choropleth, or a loan purpose automatically cross-filters every other view on the dashboard simultaneously. A dedicated grade filter panel will allow toggling individual grades globally across all charts. These linked interactions are central to the exploratory goal of the project.

Tools planned: Python with Pandas for data preprocessing and feature engineering, and Tableau for interactive visualizations. Tableau was chosen for its native support for linked dashboard interactions, built-in choropleth mapping for US states, and rapid iteration capability. Preprocessed data will be exported from Python as a CSV and loaded directly into Tableau.

Feature Engineering

The most informative features for grade prediction and visualization are expected to be: `int_rate` (assigned interest rate, a direct proxy for perceived risk), `dti` (debt-to-income ratio), `fico_range_low` (lower bound of the borrower's FICO credit score), `revol_util` (revolving credit utilization as a percentage), `annual_inc` (self-reported annual income), `loan_amnt` (requested loan size), `purpose` (categorical reason for borrowing), `addr_state` (borrower's US state), and `issue_d_year` (year of loan origination). These provide a good mix of quantitative, categorical, and geographic attributes and cover the key dimensions of credit risk.

Some derived features may be added, such as the age of the oldest credit line (computed from `earliest_cr_line_year` and `issue_d_year`), which is a common proxy for credit maturity.

Predictive Modeling (Optional)

As an optional extension, we plan to train a classifier to predict the loan grade from borrower attributes. Since `grade` is an ordinal variable with seven classes (A through G), the task is a

multi-class classification problem. The ordinal nature of the target means that misclassifying a grade-A loan as grade-C is more costly than misclassifying it as grade-B, which may justify using an ordinal classification approach rather than standard multi-class treatment. A gradient boosting model such as XGBoost or LightGBM is a natural fit given the mix of numerical and categorical features and the class imbalance toward grades A and B.

The learning set will be used for training and cross-validation, with stratified k-fold splits to account for the imbalanced class distribution. The test set will be used for final predictions in the submission format defined by `loan-10k_sol_ex.csv`. Model performance will be evaluated using weighted F1-score to handle class imbalance fairly across all grade levels.

Feature importances from the trained model will also feed back into the visualization layer, helping to identify which attributes are most predictive and therefore most worth exploring visually. Feature importance outputs will be exported from Python as a CSV and visualized as a ranked bar chart in Tableau, connecting the modeling layer back to the visualization dashboard.

Expected Outcomes

We expect to find that a small set of financial indicators, primarily FICO score, interest rate, and debt-to-income ratio, account for most of the variation in loan grade, while categorical factors such as loan purpose introduce meaningful secondary patterns. Geographic analysis is expected to reveal regional clustering, with certain states consistently producing higher-risk loan profiles. Temporally, we anticipate a shift in grade composition over the 2012 to 2018 period that reflects broader changes in lending standards.

The final deliverables will include a set of linked interactive visualizations covering the financial, geographic, and temporal dimensions of the data, accompanied by a short written summary of the key findings. Together, these are intended to give a reader a clear and honest picture of what drives credit risk in this dataset.